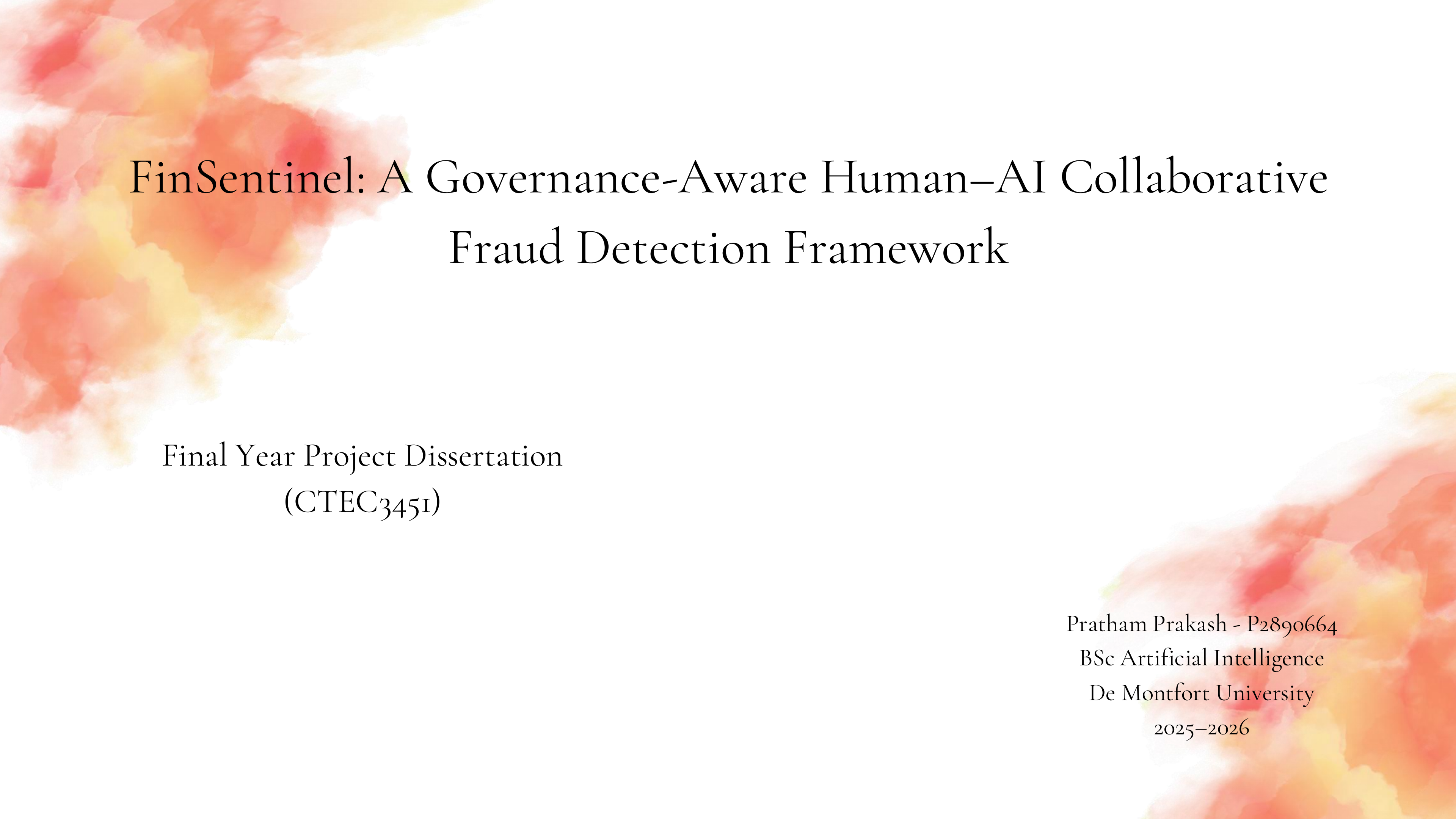




FinSentinel: A Governance-Aware Human–AI Collaborative Fraud Detection Framework

Final Year Project Dissertation
(CTEC_{345I})

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Problem Statement & Research Gap

Current Challenges in Fraud Detection

- Black-box decision behaviour
- Limited operational governance
- Limited human oversight
- Evolving fraud behaviour
- Limited long-term reliability

Research Gap

Existing approaches often investigate:

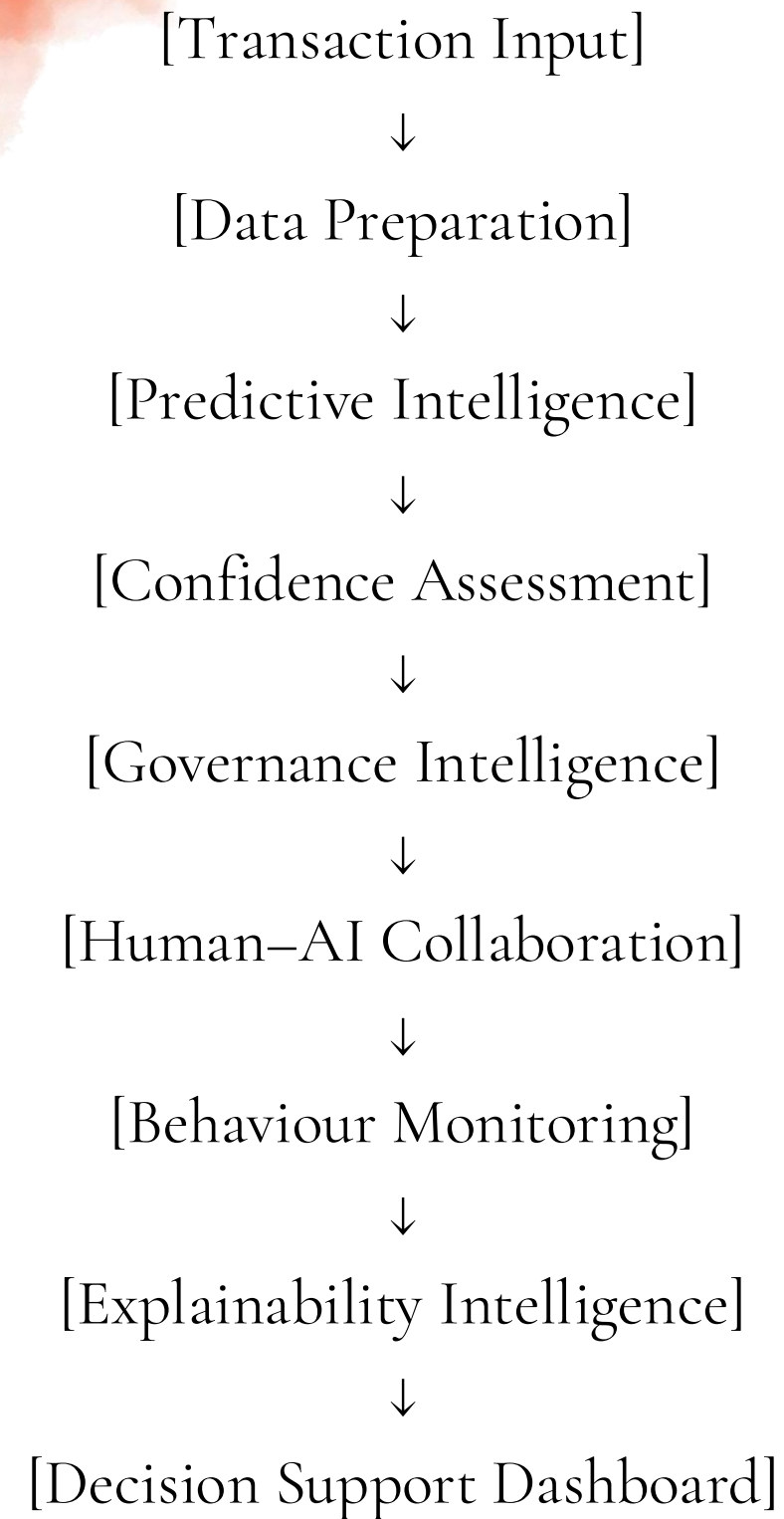
- Predictive models
- Explainability methods
- Human–AI collaboration
- Drift detection

Independently rather than within a unified framework

Project Aim:

Develop a governance-aware explainable Human–AI collaborative fraud detection framework integrating: Predictive Intelligence, Operational Routing, Human Oversight, Behavioural Drift Intelligence

FinSentinel Framework Architecture



Integrated Components

- Predictive Intelligence
- Governance Intelligence
- Explainability Intelligence
- Behavioural Monitoring
- Human Oversight

Dataset & Data Preparation

Dataset

- Credit Card Fraud Detection Dataset - Kaggle
- Total Transactions: 284,807
- Fraud Cases: 492
- Highly imbalanced dataset

Preprocessing Pipeline

- Missing value inspection
- Numeric feature filtering
- Leakage detection
- Temporal ordering using Time variable
- Train / Validation / Test split
- StandardScaler (LR & MLP only)
- Weighted Scaling (Scale_pro_weights)

Standardisation

$$z = (x - \mu) / \sigma$$

where:

- x = feature value
- μ = mean
- σ = standard deviation

Predictive Intelligence Implementation

Implemented Models

Classical Model

- Logistic Regression

Ensemble Models

- Random Forest
- XGBoost
- LightGBM
- CatBoost

Neural Model

- Multi-Layer Perceptron (MLP)

Model Evaluation Metrics

- Precision
- Recall
- F1-score
- PR-AUC
- ROC-AUC
- Brier Score*

Prediction Output

Input Features



Machine Learning Model



Fraud Probability

Selected Operational Model - Random Forest

Reason:

Strong predictive performance, Robust behaviour under imbalanced data, Suitable for operational deployment

Confidence Assessment

Probability Calibration

Calibration:

- Method = Sigmoid (Platt Scaling)
- Cross Validation = 3-fold

Purpose:

- Improve probability reliability
- Align predicted confidence with observed outcomes
- Support downstream governance decisions

Threshold Optimisation

Implemented using:

- Threshold range: 0.01–0.99
- Total thresholds evaluated: 1000
- Selection criterion: Maximum F1-score

$$F1 = (2 \times (Precision \times Recall) \div (Precision + Recall))$$

Threshold Selection Formula Model Training

For each threshold t:

Probability → Prediction



Calculate F1 score



Store F1(t)



Best Threshold
 $t^* = \text{argmax}(F1(t))$

Threshold search:

0.01–0.99

(1000 candidate thresholds)

Governance Intelligence & Human–AI Collaboration

Governance Strategy Configuration

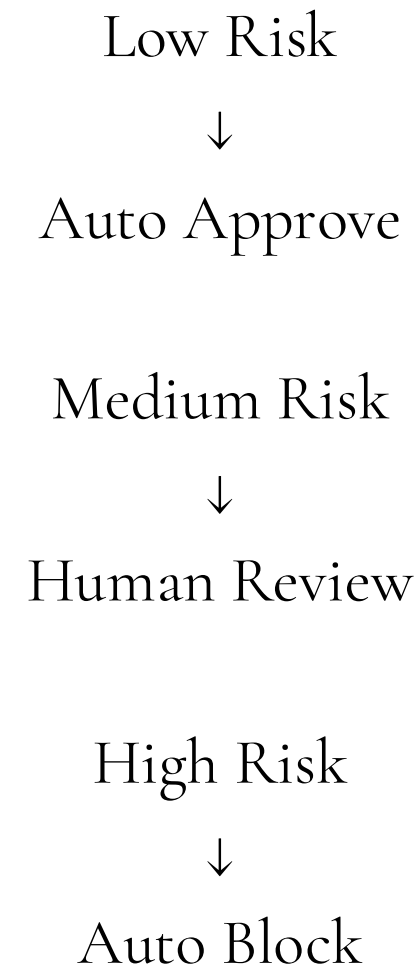
Purpose:

- Convert prediction confidence into operational decisions
- Balance fraud detection and analyst workload
- Support different governance strategies

Strategies:

- Balanced Governance
- Conservative Governance
- Aggressive Fraud Containment
- Operational Efficiency

Human–AI Routing Logic



Routing Logic

```
if p < approve_threshold:  
    Auto Approve  
  
elif p >= block_threshold:  
    Auto Block  
  
else:  
    Human Review
```

Behavioural Drift Intelligence

Drift Detection Method

Algorithm:

Page–Hinkley

Purpose:

- Monitor changing transaction behaviour
- Detect behavioural instability
- Support long-term operational reliability

Parameters Used:

- min_instances = 50 - ignore
- delta = 0.02 - max detectable change
- threshold = 50 - accumulated evidence

Adaptive Governance Response

Low Drift

No threshold adjustment

Moderate Drift

Threshold Adjustment = -0.05

High Drift

Threshold Adjustment = -0.10

Behavioural Monitoring Workflow

Transaction Behaviour



Page–Hinkley Detection



Drift Score



Governance Adjustment

Explainability Intelligence

SHAP Integration

Purpose:

- Improve prediction transparency
- Explain model behaviour
- Support analyst interpretation
- Increase operational trust

Explainers Used

- Tree Models - TreeExplainer
- Linear Models - LinearExplainer
- Neural Models - KernelExplainer

Explainability Outputs

- Global Feature Importance
- Local Prediction Explanations
- SHAP Waterfall Analysis
- Feature Contribution Analysis
- Analyst Narrative Support

SHAP Prediction Logic

Final Prediction = Base Value + Σ Feature Contributions



Unified Dashboard & Decision Intelligence

Integrated Dashboard Modules

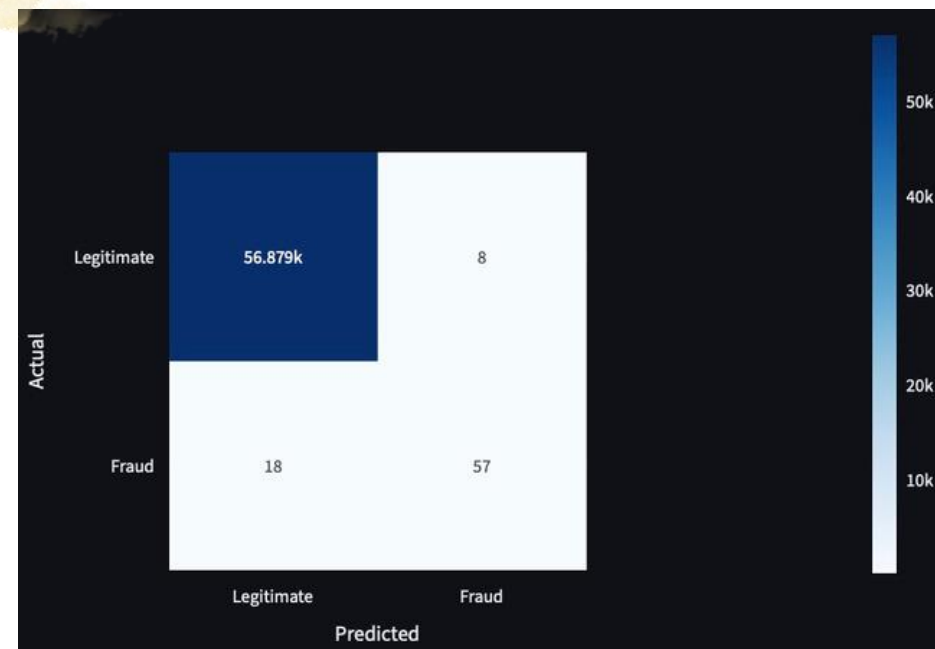
- Predictive Intelligence
- Governance Threshold Experimentation
- Explainability Outputs
- Behavioural Drift Monitoring
- Operational Analytics

Decision Intelligence Functions

- Compare AI predictions with ground truth
 - Compare Human–AI governance outcomes
 - Evaluate operational impact
 - Support threshold experimentation
 - Analyse analyst workload
- 

Results & Evaluation

AI Confusion Matrix



Governance Routing Matrix



Key Findings

Selective escalation improved fraud handling while introducing operational cost trade-offs

Human review reduced missed fraud risk

Wrongful auto-blocking remained minimal

Trade-off observed between fraud capture and operational cost

Increased intervention → increased analyst workload

Metric	AI Only	Human-AI
Fraud detected	57	72
Fraud missed	18	3
Human intervention	0	8420



Research Question Evaluation

	Research Q
RQ1:	Can governance-aware AI improve fraud detection
RQ2:	Can selective escalation reduce false positives
RQ3:	Can explainability improve user trust

Overall Finding:

FinSentinel successfully addressed all research objectives.



Contributions & Future Work

Project Contributions

- Human–AI collaborative governance framework
- Integration of predictive intelligence, explainability and behavioural monitoring
- Threshold-based governance routing for operational decision support
- Probability calibration for confidence reliability
- Unified dashboard for operational visualisation

Key Contribution

FinSentinel extends beyond prediction-focused systems by combining predictive intelligence with governance, explainability, behavioural monitoring and human oversight.

Future Work

- Real-time transaction streaming
- Dynamic threshold adaptation
- Analyst feedback learning
- Reinforcement-based decision adaptation
- Large Language Model integration
- More extensive hyperparameter optimisation across predictive models
- Learnable governance strategy optimisation rather than fixed configurations
- Adaptive drift response using data-driven threshold adjustment
- Integrated calibration-aware decision optimisation
- Reinforcement learning for dynamic governance adaptation



THANK YOU!



MODEL RUN & QUESTIONS*